

Sample configuration

In this example, both the FortiGate unit and the Cisco 2950 switch are installed and connected and basic configuration has been completed. On the switch, you need access to the CLI to enter commands. No VDOMs are enabled in this example.

General configuration steps include:

1. [Configure the external interface.](#)
2. [Add two VLAN subinterfaces to the internal network interface.](#)
3. [Add firewall addresses and address ranges for the internal and external networks.](#)
4. [Add security policies](#) to allow:
 - the VLAN networks to access each other.
 - the VLAN networks to access the external network.

To configure the external interface:

```
config system interface
  edit external
    set mode static
    set ip 172.16.21.2 255.255.255.0
  next
end
```

To add VLAN subinterfaces:

```
config system interface
  edit VLAN_100
    set vdom root
    set interface internal
    set type vlan
    set vlanid 100
    set mode static
    set ip 10.1.1.1 255.255.255.0
    set allowaccess https ping
  next
  edit VLAN_200
    set vdom root
    set interface internal
    set type vlan
    set vlanid 200
    set mode static
    set ip 10.1.2.1 255.255.255.0
    set allowaccess https ping
  next
end
```

To add the firewall addresses:

```
config firewall address
  edit VLAN_100_Net
    set type ipmask
    set subnet 10.1.1.0 255.255.255.0
  next
  edit VLAN_200_Net
    set type ipmask
    set subnet 10.1.2.0 255.255.255.0
  next
end
```

To add security policies:

Policies 1 and 2 do not need NAT enabled, but policies 3 and 4 do need NAT enabled.

```
config firewall policy
  edit 1
    set srcintf VLAN_100
    set srcaddr VLAN_100_Net
    set dstintf VLAN_200
    set dstaddr VLAN_200_Net
    set schedule always
    set service ALL
    set action accept
    set nat disable
    set status enable
  next
  edit 2
    set srcintf VLAN_200
    set srcaddr VLAN_200_Net
    set dstintf VLAN_100
    set dstaddr VLAN_100_Net
    set schedule always
    set service ALL
    set action accept
    set nat disable
    set status enable
  next
  edit 3
    set srcintf VLAN_100
    set srcaddr VLAN_100_Net
    set dstintf external
    set dstaddr all
    set schedule always
    set service ALL
    set action accept
    set nat enable
    set status enable
  next
  edit 4
```

```
set srcintf VLAN_200
set srcaddr VLAN_200_Net
set dstintf external
set dstaddr all
set schedule always
set service ALL
set action accept
set nat enable
set status enable
next
end
```

VLANs in transparent mode

In transparent mode, the FortiGate unit behaves like a layer-2 bridge but can still provide services such as antivirus scanning, web filtering, spam filtering, and intrusion protection to traffic. Some limitations of transparent mode is that you cannot use Agentless VPN, PPTP/L2TP VPN, DHCP server, or easily perform NAT on traffic. The limits in transparent mode apply to IEEE 802.1Q VLAN trunks passing through the unit.

You can insert the FortiGate unit operating in transparent mode into the VLAN trunk without making changes to your network. In a typical configuration, the FortiGate unit internal interface accepts VLAN packets on a VLAN trunk from a VLAN switch or router connected to internal network VLANs. The FortiGate external interface forwards VLAN-tagged packets through another VLAN trunk to an external VLAN switch or router and on to external networks such as the Internet. You can configure the unit to apply different policies for traffic on each VLAN in the trunk.

To pass VLAN traffic through the FortiGate unit, you add two VLAN subinterfaces with the same VLAN ID, one to the internal interface and the other to the external interface. You then create a security policy to permit packets to flow from the internal VLAN interface to the external VLAN interface. If required, create another security policy to permit packets to flow from the external VLAN interface to the internal VLAN interface. Typically in transparent mode, you do not permit packets to move between different VLANs. Network protection features such as spam filtering, web filtering, and anti-virus scanning, are applied through the UTM profiles specified in each security policy, enabling very detailed control over traffic.

When the FortiGate unit receives a VLAN-tagged packet on a physical interface, it directs the packet to the VLAN subinterface with the matching VLAN ID. The VLAN tag is removed from the packet and the FortiGate unit then applies security policies using the same method it uses for non-VLAN packets. If the packet exits the FortiGate unit through a VLAN subinterface, the VLAN ID for that subinterface is added to the packet and the packet is sent to the corresponding physical interface.

Sample topology

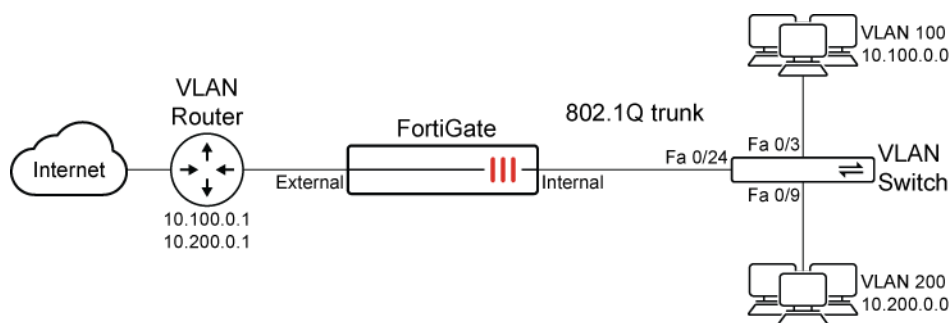
In this example, the FortiGate unit is operating in transparent mode and is configured with two VLANs: one with an ID of 100 and the other with ID 200. The internal and external physical interfaces each have two VLAN subinterfaces, one for VLAN_100 and one for VLAN_200.

The IP range for the internal VLAN_100 network is 10.100.0.0/255.255.0.0, and for the internal VLAN_200 network is 10.200.0.0/255.255.0.0.

The internal networks are connected to a Cisco 2950 VLAN switch which combines traffic from the two VLANs onto one in the FortiGate unit's internal interface. The VLAN traffic leaves the FortiGate unit on the external

network interface, goes on to the VLAN switch, and on to the Internet. When the FortiGate unit receives a tagged packet, it directs it from the incoming VLAN subinterface to the outgoing VLAN subinterface for that VLAN.

In this example, we create a VLAN subinterface on the internal interface and another one on the external interface, both with the same VLAN ID. Then we create security policies that allow packets to travel between the VLAN_100_int interface and the VLAN_100_ext interface. Two policies are required: one for each direction of traffic. The same is required between the VLAN_200_int interface and the VLAN_200_ext interface, for a total of four security policies.



Sample configuration

There are two main steps to configure your FortiGate unit to work with VLANs in transparent mode:

1. [Add VLAN subinterfaces.](#)
2. [Add security policies.](#)

You can also configure the protection profiles that manage antivirus scanning, web filtering, and spam filtering.

To add VLAN subinterfaces:

```

config system interface
  edit VLAN_100_int
    set type vlan
    set interface internal
    set vlanid 100
  next
  edit VLAN_100_ext
    set type vlan
    set interface external
    set vlanid 100
  next
  edit VLAN_200_int
    set type vlan
    set interface internal
    set vlanid 200
  next
  edit VLAN_200_ext
    set type vlan
    set interface external
    set vlanid 200

```

```
    next
end
```

To add security policies:

```
config firewall policy
    edit 1
        set srcintf VLAN_100_int
        set srcaddr all
        set dstintf VLAN_100_ext
        set dstaddr all
        set action accept
        set schedule always
        set service ALL
    next
    edit 2
        set srcintf VLAN_100_ext
        set srcaddr all
        set dstintf VLAN_100_int
        set dstaddr all
        set action accept
        set schedule always
        set service ALL
    next
    edit 3
        set srcintf VLAN_200_int
        set srcaddr all
        set dstintf VLAN_200_ext
        set dstaddr all
        set action accept
        set schedule always
        set service ALL
    next
    edit 4
        set srcintf VLAN_200_ext
        set srcaddr all
        set dstintf VLAN_200_int
        set dstaddr all
        set action accept
        set schedule always
        set service ALL
    next
end
```

Virtual VLAN switch

The hardware switch ports on FortiGate models that support virtual VLAN switches can be used as a layer 2 switch. Virtual VLAN switch mode allows 802.1Q VLANs to be assigned to ports, and the configuration of one interface as a trunk port.

The following FortiGate series are supported in FortiOS 8.0: 40F, 50G, 60F, 70F, 70G, 80F, 90G, 100F, 120G, 200F, 200G, 300E, 400E, 400F, 600F, 700G, 900G, 1100E, 1800F, 2600F, 3000F, 3200F, 3500F, 3700F, 4200F, 4400F, and 4800F. FortiWiFi 60F models are not supported.

The `virtual-switch-vlan` option must be enabled in the CLI to configure VLAN switch mode from the GUI or CLI.

To enable VLAN switches:

```
config system global
    set virtual-switch-vlan enable
end
```

After this setting is enabled, any previously configured hardware switches will appear in the *Network > Interfaces* page under *VLAN Switch*.

To enable VLAN switch mode in the GUI:

1. Go to *System > Settings > General*.
2. In the *System operation settings* section, enable *VLAN switch mode*.
3. Click *Apply*.

Basic configurations

Hardware switch ports can be configured as either a VLAN switch port or a trunk port. The available interfaces and allowable VLAN IDs that can be used depend on the FortiGate model. It is recommended to remove ports from the default VLAN switch before you begin configurations.

To create a new VLAN and assign ports in the GUI:

1. Go to *Network > Interfaces* and click *Create New > Interface*.
2. Enter a name and configure the following:
 - a. Set the *Type* to *VLAN Switch*.
 - b. Enter a *VLAN ID*.
 - c. Click the *+* and add the *Interface Members*.
 - d. Configure the *Address* and *Administrative Access* settings as needed.
3. Click *OK*.

To create a new VLAN and assign ports in the CLI:

1. Configure the VLAN:

```
config system virtual-switch
    edit "VLAN10"
        set physical-switch "sw0"
        set vlan 10
    config port
        edit "internal1"
        next
```

```

        edit "internal2"
        next
    end
next
end

```

2. Configure the VLAN switch interface addressing:

```

config system interface
    edit "VLAN10"
        set vdom "root"
        set ip 192.168.10.99 255.255.255.0
        set allowaccess ping https ssh snmp http fgfm
        set type hard-switch
    next
end

```

To designate an interface as a trunk port:

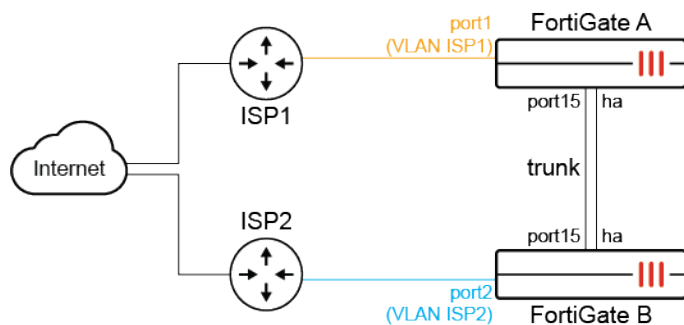
```

config system interface
    edit internal5
        set trunk enable
    next
end

```

Example 1: HA using a VLAN switch

In this example, two FortiGates in an HA cluster are connected to two ISP routers. Instead of connecting to external L2 switches, each FortiGate connects to each ISP router on the same hardware switch port on the same VLAN. A trunk port connects the two FortiGates to deliver the 802.1Q tagged traffic to the other. A full mesh between the FortiGate cluster and the ISP routers is achieved where no single point of failure will cause traffic disruptions.



This example assumes that the HA settings are already configured. The interface and VLAN switch settings are identical between cluster members and synchronized. See [HA using a hardware switch to replace a physical switch on page 4375](#) for a similar example that does not use a VLAN switch.

To configure the VLAN switches:

1. Configure the ISP interfaces with the corresponding VLAN IDs:

```
config system virtual-switch
  edit "ISP1"
    set physical-switch "sw0"
    set vlan 2951
    config port
      edit "port1"
      next
    end
  next
  edit "ISP2"
    set physical-switch "sw0"
    set vlan 2952
    config port
      edit "port2"
      next
    end
  next
end
```

2. Configure the VLAN switch interface addressing:

```
config system interface
  edit "ISP1"
    set vdom "root"
    set ip 192.168.10.99 255.255.255.0
    set allowaccess ping
    set type hard-switch
  next
  edit "ISP2"
    set vdom "root"
    set ip 192.168.20.99 255.255.255.0
    set allowaccess ping
    set type hard-switch
  next
end
```

3. Designate port15 as the trunk port:

```
config system interface
  edit port15
    set trunk enable
  next
end
```

4. Configure firewall policies to allow outgoing traffic on the ISP1 and ISP2 interfaces:

```
config firewall policy
  edit 1
    set srcintf "port11"
```



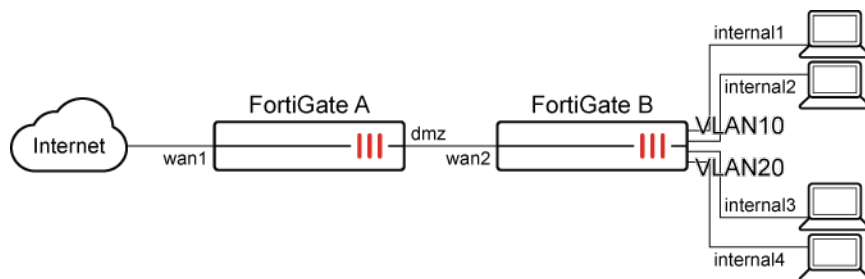
```

set dstintf "ISP1"
set srcaddr "all"
set dstaddr "all"
set action accept
set schedule "always"
set service "ALL"
set nat enable
next
edit 2
set srcintf "port11"
set dstintf "ISP2"
set srcaddr "all"
set dstaddr "all"
set action accept
set schedule "always"
set service "ALL"
set nat enable
next
end

```

Example 2: LAN extension

In this example, two hardware switch ports are assigned VLAN10, and two ports are assigned VLAN20 on FortiGate B. The wan2 interface is designated as the trunk port, and is connected to the upstream FortiGate A. The corresponding VLAN subinterfaces VLAN10 and VLAN20 on the upstream FortiGate allow further access to other networks.



The available interfaces and VLAN IDs varies between FortiGate models. The FortiGate B in this example is a 60F model.

To configure FortiGate B:

1. Configure the VLAN interfaces:

```

config system virtual-switch
edit "VLAN10"
set physical-switch "sw0"
set vlan 10
config port
edit "internal1"
next

```

```
        edit "internal2"
        next
    end
next
edit "VLAN20"
    set physical-switch "sw0"
    set vlan 20
    config port
        edit "internal3"
        next
        edit "internal4"
        next
    end
next
end
```

2. Configure the VLAN switch interface addressing:

```
config system interface
    edit "VLAN10"
        set vdom "root"
        set ip 192.168.10.99 255.255.255.0
        set allowaccess ping https ssh snmp http fgfm
        set type hard-switch
    next
    edit "VLAN20"
        set vdom "root"
        set ip 192.168.20.99 255.255.255.0
        set allowaccess ping https ssh snmp http fgfm
        set type hard-switch
    next
end
```

3. Designate wan2 as the trunk port:

```
config system interface
    edit wan2
        set trunk enable
    next
end
```

To configure FortiGate A:

1. Configure the VLAN subinterfaces:

```
config system interface
    edit "VLAN10"
        set ip 192.168.10.98 255.255.255.0
        set allowaccess ping https ssh
        set role lan
        set interface "dmz"
        set vlanid 10
```

```
next
edit "VLAN20"
    set ip 192.168.20.98 255.255.255.0
    set allowaccess ping https ssh
    set role lan
    set interface "dmz"
    set vlanid 20
next
end
```

2. Configure the DHCP server on VLAN10:

```
config system dhcp server
    edit 0
        set dns-service default
        set default-gateway 192.168.10.98
        set netmask 255.255.255.0
        set interface "VLAN10 "
        config ip-range
            edit 1
                set start-ip 192.168.10.100
                set end-ip 192.168.10.254
            next
        end
        set timezone-option default
    next
end
```

3. Configure firewall policies that allow traffic from the VLAN10 and VLAN20 interfaces to the internet:

```
config firewall policy
    edit 0
        set name "VLAN10-out"
        set srcintf "VLAN10"
        set dstintf "wan1"
        set srcaddr "all"
        set dstaddr "all"
        set action accept
        set schedule "always"
        set service "ALL"
        set nat enable
    next
    edit 0
        set name "VLAN20-out"
        set srcintf "VLAN20"
        set dstintf "wan1"
        set srcaddr "all"
        set dstaddr "all"
        set action accept
        set schedule "always"
        set service "ALL"
        set nat enable
```

```

    next
end

```

To test the connection:

1. Connect a PC to internal1 on FortiGate B.
2. Verify that it receives an IP address from FortiGate A's DHCP server.
3. From the PC, ping FortiGate B on 192.168.10.99.
4. Ping FortiGate A on 192.168.10.98.
5. Connect to the internet. Traffic is allowed by the VLAN10-out policy.

QinQ 802.1Q in 802.1ad

QinQ (802.1ad) allows multiple VLAN tags to be inserted into a single frame, and can be configured on supported FortiGate devices.

In this example, the customer connects to a provider that uses 802.1ad double-tagging to separate their customer VLANs. The FortiGate connecting to the provider double-tags its frames with an outer provider-tag (S-Tag) and an inner customer-tag (C-Tag).

Preamble & SoF 8-byte	Destination MAC 6-byte	Source MAC 6-byte	VLAN S-Tag (EtherType 0x88a8) 4-byte	VLAN C-Tag (EtherType 0x8810) 4-byte	Ether- Type 2-byte	Payload	CRC 4-byte
--------------------------	---------------------------	----------------------	---	---	--------------------------	---------	---------------

The customer identifies itself with the provider-tag (S-Tag) 232 and uses the customer-tag (C-Tag) 444 for traffic to its VLAN.



QinQ is not supported on top of hardware switch interfaces.

To configure the interfaces:

1. Configure the interface to the provider that uses the outer tag (S-Tag):

```
config system interface
  edit "vlan-8021ad"
    set vdom "root"
    set vlan-protocol 8021ad
    set device-identification enable
    set role lan
    set snmp-index 47
    set interface "PORT"
    set vlanid 232
  next
end
```

- 2. Configure a dynamic VLAN interface that uses the inner tag (C-Tag):**

```
config system interface
edit "DVLAN"
```

```

        set vdom "vdom1"
        set device-identification enable
        set role lan
        set snmp-index 48
        set interface "vlan-8021ad"
        set vlanid 444
    next
end
```

QinQ 802.1Q in 802.1Q

QinQ (802.1Q in 802.1Q) is supported for FortiGate VM models, where multiple VLAN tags can be inserted into a single frame.

Preamble & SoF 8-byte	Destination MAC 6-byte	Source MAC 6-byte	VLAN Outer Tag (Ether- Type 0x8100) 4-byte	VLAN Inner Tag (Ether- Type 0x8100) 4-byte	Ether- Type 2-byte	Payload	CRC 4-byte
--------------------------	---------------------------	----------------------	---	---	--------------------------	---------	---------------

In this example, the FortiGate VM is connected to a provider vSwitch and then a customer switch. The FortiGate encapsulates the frame with an outer 802.1Q tag of VLAN 100 and an inner 802.1Q tag of VLAN 200; port5 is used as the physical port. The provider vSwitch strips the outer tag and forwards traffic to the appropriate customer. Then the customer switch strips the inner tag and forwards the packet to the appropriate customer VLAN.



QinQ is not supported on top of hardware switch interfaces.

To configure the interfaces:

1. Configure the interface to the provider that uses the outer tag:

```
config system interface
    edit "vlan-8021q"
        set vdom "root"
        set device-identification enable
        set role lan
        set interface "port5"
        set vlan-protocol 8021q
        set vlanid 100
    next
end
```

- 2. Configure the interface to the provider that uses the inner tag:**

```
config system interface
  edit "vlan-qinq8021q"
    set vdom "root"
```

```

        set ip 1.1.1.71 255.255.255.0
        set allowaccess ping https ssh snmp http
        set device-identification enable
        set role lan
        set interface "vlan-8021q"
        set vlanid 200
    next
end

```

To verify the traffic:

1. From the FortiGate, ping 1.1.1.72:

```

# execute ping 1.1.1.72
PING 1.1.1.72 (1.1.1.72): 56 data bytes
64 bytes from 1.1.1.72: icmp_seq=0 ttl=255 time=0.2 ms
64 bytes from 1.1.1.72: icmp_seq=1 ttl=255 time=0.1 ms
64 bytes from 1.1.1.72: icmp_seq=2 ttl=255 time=0.1 ms
64 bytes from 1.1.1.72: icmp_seq=3 ttl=255 time=0.1 ms
^C
--- 1.1.1.72 ping statistics ---
4 packets transmitted, 4 packets received, 0% packet loss
round-trip min/avg/max = 0.1/0.1/0.2 ms

```

2. Verify the packet capture frame header output captured from the FortiGate's port5:

```

Frame 2: 106 bytes on wire (848 bits), 106 bytes captured (848 bits)
Ethernet II, Src: VMware_93:ae:8f (00:50:56:93:ae:8f), Dst: VMware_93:e3:72
(00:50:56:93:e3:72)
    Destination: VMware_93:e3:72 (00:50:56:93:e3:72)
    Source: VMware_93:ae:8f (00:50:56:93:ae:8f)
    Type: 802.1Q Virtual LAN (0x8100)
802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 100
    000. .... = Priority: Best Effort (default) (0)
    ...0 .... = DEI: Ineligible
    .... 0000 0110 0100 = ID: 100
    Type: 802.1Q Virtual LAN (0x8100)
802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 200
    000. .... = Priority: Best Effort (default) (0)
    ...0 .... = DEI: Ineligible
    .... 0000 1100 1000 = ID: 200
    Type: IPv4 (0x0800)
Internet Protocol Version 4, Src: 1.1.1.71, Dst: 1.1.1.72
Internet Control Message Protocol

```

The outer tag (first tag) is an 802.1Q tag with VLAN ID 100. The inner tag (second tag) is also an 802.1Q tag with VLAN ID 200.

Aggregation and redundancy

Link aggregation (IEEE 802.3ad/802.1ax) enables you to bind two or more physical interfaces together to form an aggregated (combined) link. This new link has the bandwidth of all the links combined. If a link in the group fails, traffic is transferred automatically to the remaining interfaces. The only noticeable effect is reduced bandwidth.

This feature is similar to redundant interfaces. The major difference is a redundant interface group only uses one link at a time, where an aggregate link group uses the total bandwidth of the functioning links in the group, up to eight (or more).

An interface is available to be an aggregate interface if:

- It is a physical interface and not a VLAN interface or subinterface.
- It is not already part of an aggregate or redundant interface.
- It is in the same VDOM as the aggregated interface. Aggregate ports cannot span multiple VDOMs.
- It does not have an IP address and is not configured for DHCP or PPPoE.
- It is not referenced in any security policy, VIP, IP Pool, or multicast policy.
- It is not an HA heartbeat interface.
- It is not one of the FortiGate-5000 series backplane interfaces.

When an interface is included in an aggregate interface, it is not listed on the *Network > Interfaces* page. Interfaces still appear in the CLI although configuration for those interfaces do not take affect. You cannot configure the interface individually and it is not available for inclusion in security policies, VIPs, IP pools, or routing.

Example configuration

This example creates an aggregate interface on a FortiGate-140D POE using ports 3-5 with an internal IP address of 10.1.1.123, as well as the administrative access to HTTPS and SSH.

To create an aggregate interface in the GUI:

1. Go to *Network > Interfaces* and select *Create New > Interface*.
2. Set *Name* to *aggregate*.
3. Set *Type* to *802.3ad Aggregate*.
4. Set *Interface members* to *port4*, *port5*, and *port6*.
5. Set *Addressing mode* to *Manual*.
6. Set *IP/Netmask* to *10.1.1.123/24*.
7. For *Administrative Access*, select *HTTPS* and *SSH*.
8. Click *OK*.

To create an aggregate interface in the CLI:

```
config system interface
  edit "aggregate"
    set vdom "root"
    set ip 10.1.1.123 255.255.255.0
```

```
set allowaccess https ssh
set type aggregate
set member "port4" "port5" "port6"
set snmp-index 45
next
end
```

Redundancy

In a redundant interface, traffic only goes over one interface at any time. This differs from an aggregated interface where traffic goes over all interfaces for increased bandwidth. This difference means redundant interfaces can have more robust configurations with fewer possible points of failure. This is important in a fully-meshed HA configuration.

An interface is available to be in a redundant interface if:

- It is a physical interface and not a VLAN interface.
- It is not already part of an aggregated or redundant interface.
- It is in the same VDOM as the redundant interface.
- It does not have an IP address and is not configured for DHCP or PPPoE.
- It has no DHCP server or relay configured on it.
- It does not have any VLAN subinterfaces.
- It is not referenced in any security policy, VIP, or multicast policy.
- It is not monitored by HA.
- It is not one of the FortiGate-5000 series backplane interfaces.

When an interface is included in a redundant interface, it is not listed on the *Network > Interfaces* page. You cannot configure the interface individually and it is not available for inclusion in security policies, VIPs, or routing.

Example configuration

To create a redundant interface in the GUI:

1. Go to *Network > Interfaces* and select *Create New > Interface*.
2. Set *Name* to *redundant*.
3. Set *Type* to *Redundant Interface*.
4. Set *Interface members* to *port4*, *port5*, and *port6*.
5. Set *Addressing mode* to *Manual*.
6. Set *IP/Netmask* to *10.13.101.100/24*.
7. For *Administrative Access*, select *HTTPS* and *SSH*.
8. Click *OK*.

To create a redundant interface in the CLI:

```
config system interface
edit "redundant"
```



```

    set vdom "root"
    set ip 10.13.101.100 255.255.255.0
    set allowaccess https http
    set type redundant
    set member "port4" "port5" "port6"
    set snmp-index 9
next
end

```

Enhanced hashing for LAG member selection

FortiGate models that have an internal switch that supports modifying the distribution algorithm can use enhanced hashing to help distribute traffic evenly, or load balance, across links on the Link Aggregation (LAG) interface.

The enhanced hashing algorithm is based on a 5-tuple of the IP protocol, source IP address, destination IP address, source port, and destination port.

Different computation methods allow for more variation in the load balancing distribution, in case one algorithm does not distribute traffic evenly between links across different XAUIs. The available methods are:

xor16	Use the XOR operator to make a 16 bit hash.
xor8	Use the XOR operator to make an 8 bit hash.
xor4	Use the XOR operator to make a 4 bit hash.
crc16	Use the CRC-16-CCITT polynomial to make a 16 bit hash.



The following NP6 non-service FortiGate models support this feature: 1500D, 1500DT, 3000D, 3100D, 3200D, 3700D, and 5001D.

To configure the enhanced hashing:

```

config system npu
  set lag-out-port-select {enable | disable}
  config sw-eh-hash
    set computation {xor4 | xor8 | xor16 | crc16}
    set ip-protocol {include | exclude}
    set source-ip-upper-16 {include | exclude}
    set source-ip-lower-16 {include | exclude}
    set destination-ip-upper-16 {include | exclude}
    set destination-ip-lower-16 {include | exclude}
    set source-port {include | exclude}
    set destination-port {include | exclude}
    set netmask-length {0 - 32}
  end
end

```

For example, to use XOR16 and include all of the fields in the 5-tuple to compute the link in the LAG interface that the packet is distributed to:

```

config system npu
    set lag-out-port-select enable
    config sw-eh-hash
        set computation xor16
        set ip-protocol include
        set source-ip-upper-16 include
        set source-ip-lower-16 include
        set destination-ip-upper-16 include
        set destination-ip-lower-16 include
        set source-port include
        set destination-port include
        set netmask-length 32
    end
end

```

LAG interface status signals to peer device

FortiGate can signal LAG (link aggregate group) interface status to the peer device. If the number of available links in the LAG on the FortiGate falls below the configured minimum number of links (`min-links`), the LAG interface goes down on both the FortiGate and the peer device.

When the minimum number of links is satisfied again, the LAG interface automatically resumes operation on both the FortiGate and the peer device. While the LAG interface is down, interface members are in the Link Aggregation Control Protocol (LACP) MUX state of *Waiting*.

Example

In this example, the LAG interface is configured on FGT_A and peered with FGT_B.

To verify the configuration:

1. On FGT_A, check the minimum number of links for the LAG interface named `test_agg1`.
In the following example, set `min-links 1` indicates that a minimum of one alive interface member is required to keep the LAG interface up.

```

# show
config system interface
    edit "test_agg1"
        set vdom "vdom1"
        set ip 11.1.1.1 255.255.255.0
        set allowaccess ping https
        set type aggregate
        set member "port7" "port8" "port9"
        set device-identification enable
        set lldp-transmission enable
        set role lan
        set snmp-index 41
        set min-links 1
    next
end

```